SIMATS ENGINEERING

ARTIFICIAL INTELLIGENCE – CO4 ASSESSMENT TOOL 2

Assessment	Article Review – Annexure A
Selected Topic	Decision Trees / Classification
Course Name	Artificial Intelligence
Course Code	CSA17
Course Outcome	CO4 – Implement machine learning techniques including decision tree algorithms, reinforcement learn ng, and buil
Name	Tharish K
Register No.	192412004

Selected Article
Moslehi, S., Rabiei, N., Soltanian, A. R., & Mamani, M. (2022). *Application of machine learning models based on decision trees in classifying the factors affecting mortality of COVID-19 patients in Hamadan, Iran.* BMC Medical Informatics and Decision Making, 22, 192.

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Springer article: https://link.springer.com/article/10.1186/s12911-022-01939-x

TASK 1: ARTICLE SUMMARY

1(a) Full Citation, Domain and ML Problem

Full Citation: Moslehi, S., Rabiei, N., Soltanian, A. R., & Mamani, M. (2022). Application of machine learning models based on decision trees in classifying the factors affecting mortality of COVID-19 patients in Hamadan, Iran. *BMC Medical Informatics and Decision Making*, 22, 192. DOI: 10.1186/s12911-022-01939-x.

Domain/Application: Medical informatics and clinical decision support.

ML Problem: Binary classification of hospitalized COVID-19 patient outcomes into recovered or deceased, while identifying demographic, clinical and laboratory factors that contribute most strongly to mortality.

1(b) Article Objective and Research Gap (150–200 words)

The study aims to develop an accurate and interpretable machine-learning approach for classifying mortality among hospitalized COVID-19 patients. The authors identified a practical research gap: earlier machine-learning studies could predict or classify mortality, but often provided limited interpretation of which patient characteristics were most influential and how those factors related to the final outcome. To address this gap, the researchers used a two-stage strategy. First, a random forest containing 1,000 trees was used to estimate feature importance among 25 demographic, clinical and laboratory variables. Features with relative importance above 6% were retained. These were gender, age, blood urea nitrogen (BUN), partial thromboplastin time (PTT), serum glutamic-oxaloacetic transaminase (SGOT) and erythrocyte sedimentation rate (ESR). Next, four interpretable classification models—logistic model tree (LMT), C4.5, C5.0 and CART—were developed and compared using accuracy, recall and F1-score. CART was selected as the best-performing model and its tree structure was used to interpret mortality risk. Thus, the work combines predictive classification with feature prioritization and an understandable decision structure, which is valuable for clinical decision-making.

TASK 2: METHODOLOGY ANALYSIS

2(a) ML Techniques and Dataset

Dataset: The retrospective study used records from 2,470 COVID-19 patients admitted to Sina and Besat hospitals in Hamadan, Iran, between February 2020 and July 2021. The outcome was binary: recovered (0) or dead (1). The dataset contained 1,853 recovered patients (75.02%) and 617 deceased patients (24.98%).

Data preparation: Missing and illogical/outlier records were removed after examining the missingness mechanism. Initially, 25 demographic, clinical and laboratory features were considered.

Feature selection: A random forest with 1,000 trees and the Gini criterion calculated relative feature importance. Features above 6% importance were retained: age, gender, BUN, PTT, SGOT and ESR.

Classification: Four tree-based models—LMT, C4.5, C5.0 and CART—were compared. Data were randomly divided into training and test sets, and the procedure was repeated 10 times to reduce the effect of one random split. Accuracy, recall and F1-score were calculated and averaged. The final model was additionally assessed using ROC and AUC. The analysis used Python 3.8 with scikit-learn and R 4.1.2 with the train package.

2(b) Mapping to CO4, Strength and Limitation

CO4 Topic	Connection to the Article
Decision Trees	Directly demonstrated through LMT, C4.5, C5.0 and CART for binary classification.
Inductive Learning	Models learn decision rules and relationships from labelled patient records and apply them to unseen cases.
Statistical Learning	Performance is evaluated quantitatively using accuracy, recall, F1-score and ROC-AUC.
Classification & Decision-making	The final CART model converts learned rules into interpretable mortality-risk decisions.

Methodological strength: The approach combines feature selection with interpretable decision-tree models. The resulting CART structure can be inspected as explicit rules, making the model easier for clinicians to understand than a purely black-box predictor.

Methodological limitation: The study used data from only two hospitals in one region of Iran, and some potentially useful variables such as ALP and ALT/SGPT were excluded because of missing information. This limits generalizability to other hospitals, populations and healthcare systems.

TASK 3: RESULTS & CRITICAL EVALUATION

3(a) Key Results and Model Comparison

Model	Training Accuracy	Training F1-score
LMT	76.89%	85.73%
C4.5	77.70%	86.08%
C5.0	76.89%	85.74%
CART	78.24%	86.81%

Test-set results reported for the final CART model: Accuracy = 78.24%, Recall = 95.50%, F1-score = 86.81%, and ROC-AUC = 79.5%. The high recall is important because failing to identify a patient at high risk of death can have serious clinical consequences.

3(b) Critical Evaluation

Realism of metrics: The reported accuracy of about 78% and AUC of about 0.80 are plausible for a clinical mortality-classification problem with substantial patient variability. The recall of 95.5% is particularly important for identifying high-risk patients. However, high recall alone does not establish that the model is well calibrated or that it will perform equally well in other hospitals.

Potential biases: The outcome classes are imbalanced, with recovered patients representing about three quarters of the dataset. A single regional healthcare setting may introduce population and treatment-pattern bias. Removing incomplete records can also affect representativeness. The dataset was not publicly available, which reduces independent reproducibility.

Additional experiments: External validation using hospitals from other regions or countries would strengthen the evidence. Stratified cross-validation, calibration analysis, sensitivity/specificity reporting, confidence intervals and subgroup evaluation by age and sex would make the findings more robust. A prospective validation study could test whether the model improves actual clinical decisions rather than only retrospective prediction.

3(c) Real-World Applicability

The method could support hospital triage, early risk prioritization and clinical decision support by highlighting a small set of influential patient factors. Because CART is interpretable, clinicians can inspect the rules instead of receiving only a numerical prediction. The model could be integrated into an electronic-health-record workflow and produce a risk category when relevant laboratory and demographic values are available.

Industry-scale challenges: Data must be complete, standardized and representative across hospitals. Laboratory reference ranges, coding practices and patient populations can differ between institutions. The model would require continuous monitoring for performance drift, cybersecurity and privacy protection. Ethical deployment should treat the model as decision support rather than an unquestioned replacement for qualified clinicians.

TASK 4: REFLECTION & LEARNING OUTCOME

4(a) Three Key Insights

Insight 1 – Decision-tree interpretability: I learned that a decision tree is not only a classifier but also a way of expressing the learned decision process as understandable rules. This connects directly to the CO4 topic of Decision Trees and classification-based decision-making.

Insight 2 – Feature selection improves focused modelling: The study first used random forest feature importance to reduce 25 variables to six influential features. This showed me that model building is not only about choosing an algorithm; selecting useful features can make a model simpler and easier to interpret. This connects to Inductive Learning and Statistical Learning.

Insight 3 – Metrics must match the application: Accuracy alone does not fully describe a medical classifier. Recall and F1-score are important when the cost of missing a high-risk patient is significant, while ROC-AUC provides another view of discrimination. This connects to CO4 classification and model evaluation.

4(b) Proposed Extension / Novel Experiment

Proposed experiment: Validate the CART model on a multi-hospital dataset from India and compare it with a cost-sensitive Random Forest and Gradient Boosting classifier, while keeping an interpretable CART model as the clinical baseline.

Justification and expected impact: The original study is limited to two hospitals in Hamadan, so external validation is the most useful next step. An Indian multi-centre dataset would test whether the selected factors and decision rules transfer across healthcare settings. Comparing several models would show whether the interpretability advantage of CART comes with a meaningful loss in predictive performance. Adding calibration and subgroup analysis would help identify whether predicted risk is reliable for different patient groups. If the results remain stable, the method could provide a practical and interpretable clinical decision-support tool.

ARTICLE SOURCE / VALID LINK

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Note: The selected article is a Springer/BMC peer-reviewed research article, not an IEEE Xplore paper. It satisfies the assignment's listed publisher options because Springer is explicitly included in the instructions.